

Sheet 4

$$y_i \sim N(\mu, \sigma^2).$$

a) The joint density function tells us the probability of observing some ~~random~~ set of $\{y_i\}$, given the parameters μ, σ^2 .

$$f(y_1, \dots, y_n | \mu, \sigma^2) = \prod_{i=1}^n f(y_i | \mu, \sigma^2) \quad \text{since } y_i \text{ is independent}$$

$$= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \prod_{i=1}^n \exp\left(-\frac{1}{2\sigma^2} \cdot (y_i - \mu)^2\right)$$

1a. yes, correct!

$$= (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2\right)$$

And then the log-likelihood is $\log f(y | \mu, \sigma^2)$

$$= \log\left[(2\pi\sigma^2)^{-n/2}\right] + \log[\exp(\dots)]$$

$$= -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (y_i - \mu)^2$$

$$=: \ell_{y_1, \dots, y_n}(\mu, \sigma^2)$$

b) The maximum likelihood estimator is defined as

$$(\hat{\mu}, \hat{\sigma}^2) = \arg \max_{(\mu, \sigma^2) \in (\mathbb{R}, \mathbb{R}^+)} \ell_{y_1, \dots, y_n}(\mu, \sigma^2)$$

For which the FOCs are $\frac{\partial \ell}{\partial \mu} \stackrel{!}{=} 0$; $\frac{\partial \ell}{\partial \sigma^2} \stackrel{!}{=} 0$

$$\text{so } \frac{\partial \ell}{\partial \mu} = -\frac{1}{\sigma^2} \sum (y_i - \mu) = 0 \Rightarrow \sum (y_i - \mu) = 0$$

$$\text{and } -\frac{n}{2} \frac{2\pi}{2\pi\sigma^2} + \frac{1}{2\sigma^2} \sum (y_i - \mu)^2 = 0 \Rightarrow n\sigma^2 = \sum (y_i - \hat{\mu})^2$$

1b. yes, correct and well noticed that the MLE is biased in small samples!

Solving, from $\sum (y_i - \mu) = 0$ we have $\hat{\mu} = \bar{n} \sum y_i$,
i.e. the sample mean $\bar{y}$.

And then $\hat{\sigma}^2$ is the (biased) yes...but double-check with tutor!

$$\hat{\sigma}^2 = \frac{1}{n} \sum (y_i - \bar{y})^2$$

* Are ML estimators not necessarily unbiased? *

c) Taking the second partial derivatives, we have

$$\frac{\partial^2 \ell}{\partial \mu^2} = -\frac{n}{\sigma^2} < 0 \quad \text{Since } \sigma^2 > 0, n \in \mathbb{Z}^+$$

$$\frac{\partial^2 \ell}{\partial \sigma^2} = \frac{n}{2} (\sigma^2)^{-2} - (\sigma^2)^{-3} \sum (y_i - \mu)^2$$

under FOC, $= n \sigma^2$ at

1c. excellent, good that you calculate the cross-partial derivatives. Putting this all together, what can you say about the definiteness of the Hessian matrix?

$$\frac{\partial^2 \ell}{\partial \mu \partial \sigma^2} = (\sigma^2)^{-2} \sum (y_i - \mu)$$

under FOC, $= 0$ at optimum solution

$$= 0$$

So it's a maximum, by the second derivative test.

* Do you need to explain more here, and might you have a problem where the cross-derivative $\neq 0$? *

$$d) \quad y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad x_i \stackrel{iid}{\sim} N(0, 1) \perp \varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$$

So we have joint density function

$$f(y_i, \dots, y_n | x_i, \beta_0, \beta_1, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum (y_i - \beta_0 - \beta_1 x_i)^2\right)$$

and log-likelihood

$$\ell_{\text{JGL}}(\beta_0, \beta_1, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (y_i - \beta_0 - \beta_1 x_i)^2$$

with maximum likelihood estimators defined as

$$(\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma}^2) = \arg \max_{(\beta_0, \beta_1, \sigma^2) \in (\mathbb{R}, \mathbb{R}, \mathbb{R}^+)} \ell_{y|x}(\beta_0, \beta_1, \sigma^2)$$

for which the FOCs are

$$\left. \begin{aligned} \frac{\partial \ell}{\partial \beta_0} = 0 &\Rightarrow \frac{1}{\sigma^2} \sum (y_i - \beta_0 - \beta_1 x_i) = 0 \\ \frac{\partial \ell}{\partial \beta_1} = 0 &\Rightarrow \frac{1}{\sigma^2} \sum (y_i - \beta_0 - \beta_1 x_i) x_i = 0 \end{aligned} \right\} \begin{array}{l} \text{exactly the usual OLS} \\ \text{FOCs in the simple} \\ \text{model} \end{array}$$

$$\frac{\partial \ell}{\partial \sigma^2} = 0 \Rightarrow \sigma^2 = \frac{1}{n} \sum (y_i - \beta_0 - \beta_1 x_i)^2 \quad (\text{simplifying directly as in (b)})$$

$$s.o. \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x},$$

1d. yes, great answer!

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\text{and } \hat{\sigma}^2 = \frac{1}{n} \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

We have Normal errors ε_i independent of each other and mean-independent of the regressors, so we know that in finite samples $\hat{\beta}_0, \hat{\beta}_1$ are unbiased, BLUE_{by Gauss-Markov} and distributed Normally with variances we can compute. Moreover, the t -statistic follows a t distribution, so exact inference is possible.

In large samples, the estimators are consistent and their standardised counterparts are asymptotically normal, since we as discussed have mean-independent errors with finite conditional variance and iid sampling of (x_i, ε_i) . (Assuming x_i has nonzero variance and both have finite fourth moments)

Second-deriv tests for (a) and (b) after (c)

2a y_i iid 0 with density $f(y, \theta)$.

a) Joint density function: $\frac{1}{\theta^n} \prod_{i=1}^n e^{-y_i/\theta} = f(y_1, \dots, y_n | \theta)$
 $= \theta^{-n} \cdot \exp(-\theta^{-1} \cdot \sum y_i)$

Log-likelihood function $\ell_y(\theta) = -n \log \theta - \frac{1}{\theta} \sum y_i$

$\hat{\theta}_{ML} := \arg \max_{\theta \in \mathbb{R}} \ell_y(\theta)$ with FOC $\frac{\partial \ell}{\partial \theta} \stackrel{!}{=} 0$

2a. yes, correct!

so $-\frac{n}{\theta} + \frac{1}{\theta^2} \sum y_i = 0 \Rightarrow \hat{\theta} = \frac{1}{n} \sum y_i = \bar{y}$

b) $f(y_1, \dots, y_n | \theta) = \prod_{i=1}^n \binom{k}{y_i} \theta^{y_i} (1-\theta)^{k-y_i}$

$$= \theta^{\sum y_i} \cdot (1-\theta)^{\sum (k-y_i)} \cdot \prod_{i=1}^n \binom{k}{y_i}$$

2b. yes, correct!

$$= \theta^{n\bar{y}} \cdot (1-\theta)^{n(k-\bar{y})} \cdot \prod \binom{k}{y_i} \quad \text{with } \bar{y} := \frac{1}{n} \sum y_i$$

$$\ell_y(\theta) = n\bar{y} \log(\theta) + n(k-\bar{y}) \log(1-\theta) + \sum_{i=1}^n \log \binom{k}{y_i}$$

and $\hat{\theta}_{ML}$ is the $\arg \max$

with FOC $\frac{\partial \ell}{\partial \theta} = \frac{n\bar{y}}{\theta} - \frac{n(k-\bar{y})}{1-\theta} \stackrel{!}{=} 0$

could write as $\log k! - \log(y_i!(k-y_i)!)$

$$\Rightarrow \bar{y}(1-\hat{\theta}) = \hat{\theta}(k-\bar{y}), \text{ so then } \hat{\theta} = \bar{y}/k$$

④ Is c) $\theta = \exp(\beta) / (1 + \exp(\beta))$, $\beta \in \mathbb{R}$

There a modelling reason

to fit β rather than θ directly, imposing $\theta \in [0, 1]$?

$$f(y_1, \dots, y_n | \theta) = \theta^{n\bar{y}} \cdot (1-\theta)^{n(1-\bar{y})}$$

$$\ell_y(\theta) = n\bar{y} \log \theta + n(1-\bar{y}) \log(1-\theta)$$

$\hat{\theta}_{ML} = \bar{y}$. So we can just work backwards to then obtain $\hat{\beta}_{ML}$.

from (b) $\log \theta = \log \frac{e^\beta}{1+e^\beta}$

we have $\theta + \theta e^\beta = e^\beta$
 $\therefore e^\beta(1 - \theta) = \theta$

2c. yes, correct!

$$\beta = \log \theta - \log(1 - \theta)$$

$$\therefore \hat{\beta}_{ML} = \log \bar{y} - \log(1 - \bar{y}) = \log \frac{\bar{y}}{1 - \bar{y}}$$

And since from (b) we know $\hat{\theta}_{ML}$ was a maximum, then so too is $\hat{\beta}_{ML}$. * Is more argument required? * ($k=1$) $\rightarrow$

a) cont'd. $\frac{\partial^2 \ell}{\partial \theta^2} = n \theta^{-2} - 2 \theta^{-3} \sum y_i$

$$= n \theta^{-3} (\theta - 2 \bar{y}) \quad \text{but at optimum, } \hat{\theta} = \bar{y}$$

$$\therefore = n \hat{\theta}^{-3} (\hat{\theta} - 2 \hat{\theta}) = -n \hat{\theta}^{-2} < 0 \quad \text{since } \bar{y}; n > 0$$

So it's a maximum.

b) cont'd. $\frac{\partial^2 \ell}{\partial \theta^2} = -n \bar{y} \theta^{-2} - \frac{n(k - \bar{y})}{(1 - \theta)^2}$

which < 0 , since $\hat{\theta} \in [0, 1]$ so the denominators are +ve, and both numerators are +ve.

$$\Rightarrow \bar{y} \in [0, k]$$

So it's a maximum.

2d. yes, correct!

d) $f(y_1, \dots, y_n | \theta) = \prod_{i=1}^n \frac{1}{\theta} 1(0 \leq y_i \leq \theta) = \theta^{-n} \prod 1(0 \leq y_i \leq \theta) = L_y(\theta)$

$$\ell_y(\theta) = -n \log \theta + \sum \log 1(0 \leq y_i \leq \theta) \quad \text{where } \theta \geq \max \{y_i\}$$

Note that if $\theta < \max \{y_i\}$, we have $L_y(\theta) = 0$. And

Since the product term = 1 for $\theta \geq \max \{y_i\}$, the likelihood decreases monotonically as θ increases beyond $y_{\max}$ due to the θ^{-n} term. So we'll have $\hat{\theta}_{ML} = y_{\max}$.

3. Independent pairs (Y_i, X_i) , with $P(\text{Success}) + P(\text{Failure}) = 1$

weighted
product of
two sigmoid
and still else?
(oh, Douglas??)

$$P_{\beta_1, \beta_2}(Y_i = y_i | X_i = x_i) = \left(\frac{\exp(\beta_1 + \beta_2 x_i)}{1 + \exp(\beta_1 + \beta_2 x_i)} \right)^{y_i} \left(\frac{1}{1 + \exp(\beta_1 + \beta_2 x_i)} \right)^{1-y_i}$$

oh, it's a
Bernoulli funct

a) The likelihood function measures how well particular parameter choices for β_1, β_2 fit our observed data $\{(Y_i, X_i)\}$. Specifically, it gives us the probability (density) of observing those values given some parameters, where those parameters are arguments to the function.

This is useful because we want to find the parameters which best fit our data, so we can take the argmax as our estimators. For ease of calculation, we work with the log of the likelihood function since that transforms products into sums but preserves the optimum by monotonicity.

$$L_n(\beta_1, \beta_2) = \prod_{i=1}^n \left(\frac{e^{z_i}}{1 + e^{z_i}} \right)^{y_i} \left(\frac{1}{1 + e^{z_i}} \right)^{1-y_i} \quad \text{where } z_i = \beta_1 + \beta_2 x_i$$

$$\begin{aligned} \ell_n(\beta_1, \beta_2) &= \sum_{i=1}^n (\log e^{z_i} - \log(1 + e^{z_i})) y_i \\ &\quad + \sum_{i=1}^n (-\log(1 + e^{z_i})) (1 - y_i) \end{aligned}$$

3a. yes, correct

$$\begin{aligned} &= \sum y_i \log e^{z_i} + \sum y_i (-\log(1 + e^{z_i})) + \log(1 + e^{z_i}) \\ &\quad - \sum \log(1 + e^{z_i}) \end{aligned}$$

3b. correct!

$$= \sum y_i (\beta_1 + \beta_2 x_i) - \log(1 + e^{\beta_1 + \beta_2 x_i})$$

$$\begin{aligned} b) \quad \frac{\partial \ell}{\partial \beta_1} &= \sum y_i - \frac{e^{z_i}}{1 + e^{z_i}} \quad \text{and} \quad \frac{\partial \ell}{\partial \beta_2} = \sum y_i x_i - x_i \frac{e^{z_i}}{1 + e^{z_i}} \\ &= \sum y_i - \Lambda_i(\beta) \quad = \sum x_i (y_i - \frac{e^{z_i}}{1 + e^{z_i}}) \\ &\quad = \sum x_i (y_i - \Lambda_i(\beta)) \end{aligned}$$

c) First, note that
$$\frac{\partial \Lambda_i}{\partial \beta_1} = \frac{e^{\beta_1 + \beta_2 x_i} \cdot (1 + e^{\beta_1 + \beta_2 x_i}) - e^{\beta_1 + \beta_2 x_i} \cdot e^{\beta_1 + \beta_2 x_i}}{(1 + e^{\beta_1 + \beta_2 x_i})^2}$$

$$= \frac{e^{2x_i}}{(1 + e^{2x_i})^2}$$

and similarly,
$$\frac{\partial \Lambda_i}{\partial \beta_2} = \frac{x_i e^{2x_i} (1 + e^{2x_i}) - x_i e^{2x_i} (e^{2x_i})}{(1 + e^{2x_i})^2} = \frac{x_i e^{2x_i}}{(1 + e^{2x_i})^2}$$

So
$$H_{11} = \frac{\partial^2 \ell}{\partial \beta_1^2} = - \sum \frac{e^{2x_i}}{(1 + e^{2x_i})^2}$$

$$H_{12} = H_{21} = - \sum \frac{\partial \Lambda_i}{\partial \beta_2} = - \sum \frac{x_i e^{2x_i}}{(1 + e^{2x_i})^2}$$

by equality of mixed partials.

$$H_{22} = - \sum x_i \frac{\partial \Lambda_i}{\partial \beta_2} = - \sum \frac{x_i^2 e^{2x_i}}{(1 + e^{2x_i})^2}$$

And we thus have
$$H = - \sum_{i=1}^n \frac{e^{2x_i}}{(1 + e^{2x_i})^2} \begin{bmatrix} 1 & x_i \\ x_i & x_i^2 \end{bmatrix}$$

This is important because to show that the ~~argmax~~ solution to setting our FOCs $= 0$ is a maximum of ℓ_n , we'd need to check that H is negative definite. Also, the information matrix used to standardise $\hat{\beta}$ in the LM test is $\frac{1}{n} H$.

d)
$$\hat{\beta}_1, \hat{\beta}_2 = \underset{(\beta_1, \beta_2) \in \mathbb{R}^2}{\text{argmax}} \ell_n(\beta_1, \beta_2), \text{ which has FOCs}$$

$$\frac{\partial \ell_n}{\partial \beta_1} = \sum y_i - \Lambda_i(\beta) \stackrel{!}{=} 0, \quad \frac{\partial \ell_n}{\partial \beta_2} = \sum x_i (y_i - \Lambda_i(\beta)) \stackrel{!}{=} 0 \quad \text{from (b).}$$

We could write $\Lambda_i(\beta)$ as

$$\left(\frac{\exp(\beta_1)}{1 + \exp(\beta_1)} \right)^{1-x_i} \cdot \left(\frac{\exp(\beta_1 + \beta_2)}{1 + \exp(\beta_1 + \beta_2)} \right)^{x_i} \quad \text{since } x_i \in \{0, 1\}$$

3c. yes, correct, a complete answer would also argue why the Hessian you've derived is -ve definite

Moreover, in the sum for FOC_{β_2} , we only need to consider terms where $x_i = 1$ since the others are multiplied by zero. So the condition becomes

$$\sum y_i = \sum \frac{\exp(\beta_1 + \beta_2)}{1 + \exp(\beta_1 + \beta_2)} \cdot x_i$$

using our substitution for Λ_i and then selecting only obs where $x_i = 1$

alongside FOC_{β_1} : $\sum y_i = \sum \Lambda_i$

Exploiting the equality, we now have

$$\sum \left(\frac{\exp(\beta_1)^{1-x_i}}{1 + \exp(\beta_1)} \cdot \frac{\exp(\beta_1 + \beta_2)^{x_i}}{1 + \exp(\beta_1 + \beta_2)} \right) = \sum \frac{\exp(\beta_1 + \beta_2)}{1 + \exp(\beta_1 + \beta_2)} x_i$$

We can split the observations depending on whether $x_i = 1$ or 0.

When $x_i = 0$, y_i has probability of success $p_i = \frac{e^{\beta_1}}{1 + e^{\beta_1}} = \pi_0$. And when $x_i = 1$, $p_i = \frac{e^{\beta_1 + \beta_2}}{1 + e^{\beta_1 + \beta_2}} = \pi_1$.

The likelihood function for all our data is the product of the likelihood of the y_i 's attached to $x_i = 0$ and those attached to $x_i = 1$.
i.e. $L_n(\beta_1, \beta_2) = \pi_0^{s_0} \cdot (1 - \pi_0)^{n_0 - s_0} \cdot \pi_1^{s_1} \cdot (1 - \pi_1)^{n_1 - s_1}$

where n_k, s_k gives us the number of observations and successes in the subgroup where $x = k \in \{0, 1\}$, and $n_0 + n_1 = n$.

This is now the product of two independent Bernoulli likelihood functions, with parameters π_0, π_1 . And the MLE for π_0, π_1 will just be the empirical mean, as with θ in 2c).

3d. yes, correct!

i.e. $\hat{\pi}_0 = s_0/n_0$, $\hat{\pi}_1 = s_1/n_1$. Working backwards, then, we obtain

$\hat{\beta}_0 = \log \frac{\bar{y}_0}{1 - \bar{y}_0}$ as in 2c). And since $\hat{\beta}_1 + \hat{\beta}_2 = \log \frac{\bar{y}_1}{1 - \bar{y}_1}$, we have

$\hat{\beta}_2 = \log \frac{\bar{y}_1}{1 - \bar{y}_1} - \log \frac{\bar{y}_0}{1 - \bar{y}_0}$, where $\bar{y}_k := \frac{1}{n_k} \sum y_k$